

# Auto Shutdown AWS EC2 Using Lambda

## Introduction

Running EC2 instances for long hours can increase AWS costs, especially while practicing or working on test environments. Instead of stopping instances manually from the AWS console every time, this project shows a better and more practical approach.

In this project, I used AWS Lambda to stop running EC2 instances by triggering a Lambda function manually. This helps reduce unnecessary cloud costs and also demonstrates how serverless automation can be used for basic infrastructure management, which is commonly expected in DevOps and MNC environments.

## Requirements

AWS EC2

AWS Lambda (Python 3.10)

AWS IAM

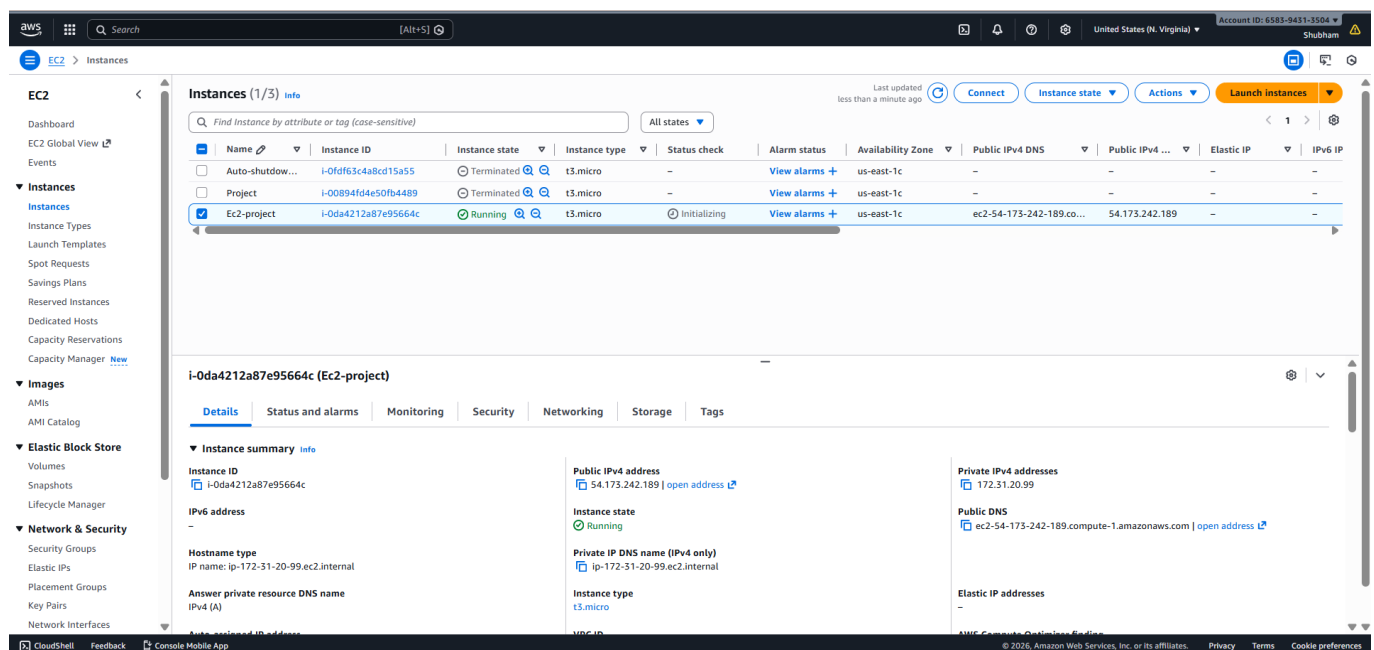
Python (boto3)

Git & GitHub

## Steps of Deployment

### step1 - Launch EC2 Instance

Create an EC2 instance using Amazon Linux to test the shutdown automation.

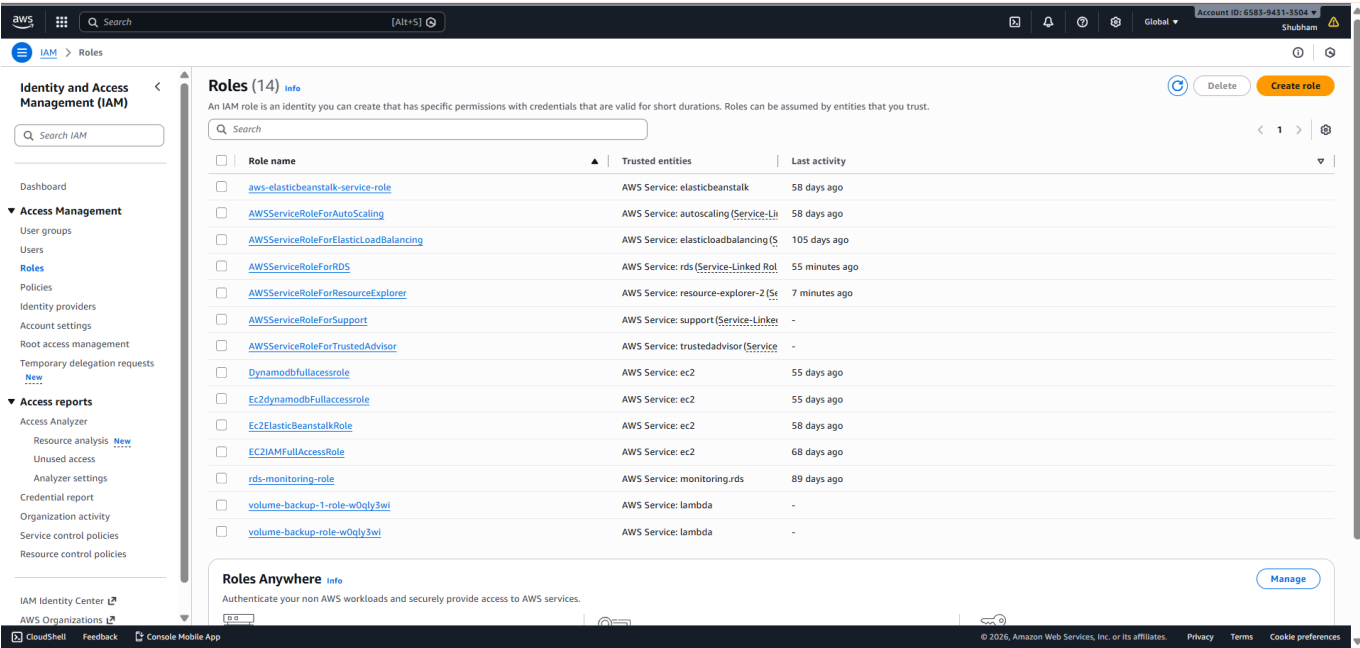


### step2 - Create IAM Role

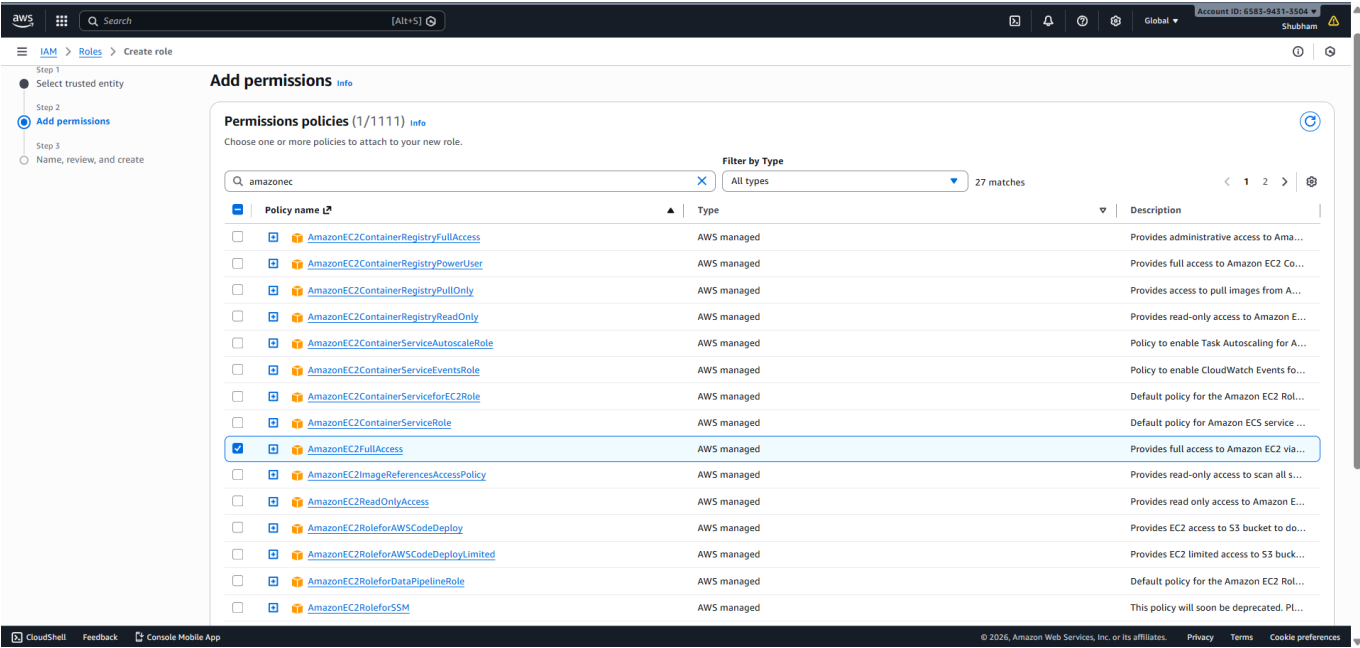
Create an IAM role for Lambda and attach the following policies:

AmazonEC2FullAccess

This role allows Lambda to stop EC2 instances and write logs.



Here, we have attached a policy to the role.



The screenshot shows the 'Name, review, and create' step in the AWS IAM console. The 'Role details' section has 'Role name' set to 'ec2-shutdown-role' and a description 'Allows Lambda functions to call AWS services on your behalf.' The 'Trust policy' section shows a JSON policy for 'Lambda.amazonaws.com'. The 'Permissions policy summary' shows 'AmazonEC2FullAccess' as the attached policy. The 'Add tags' section is empty. At the bottom, there are 'Cancel', 'Previous', and 'Create role' buttons.

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Add tags

**Name, review, and create**

**Role details**

**Role name**  
Enter a meaningful name to identify this role.

**Description**  
Add a short explanation for this role.

**Step 1: Select trusted entities**

**Trust policy**

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "sts:AssumeRole",
7       "Principal": {
8         "Service": "lambda.amazonaws.com"
9       }
10    ]
11  }
12 }
```

**Step 2: Add permissions**

**Permissions policy summary**

Policy name  Type  Attached as

**Step 3: Add tags**

**Add tags - optional** [info](#)  
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.  
No tags associated with the resource.  
[Add new tag](#)  
You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create role](#)

<https://us-east-1.console.aws.amazon.com/console/home?region=us-east-1>

Here, we have successfully created a role.

The screenshot shows the 'Roles' page in the AWS IAM console. A green banner at the top says 'Role ec2-shutdown-role created.' The table lists various roles, with 'ec2-shutdown-role' highlighted. The table has columns for 'Role name', 'Trusted entities', and 'Last activity'.

**Role ec2-shutdown-role created.**

**Roles (1/15)** [info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

| <input checked="" type="checkbox"/> | Role name                                             | Trusted entities                                        | Last activity  |
|-------------------------------------|-------------------------------------------------------|---------------------------------------------------------|----------------|
| <input type="checkbox"/>            | <a href="#">aws-elasticbeanstalk-service-role</a>     | AWS Service: elasticbeanstalk                           | 58 days ago    |
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForAutoScaling</a>          | AWS Service: autoscaling (Service-Linked Role)          | 58 days ago    |
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForElasticLoadBalancing</a> | AWS Service: elasticloadbalancing (Service-Linked Role) | 105 days ago   |
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForRD5</a>                  | AWS Service: rds (Service-Linked Role)                  | 59 minutes ago |
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForResourceExplorer</a>     | AWS Service: resource-explorer-2 (Service-Linked Role)  | 11 minutes ago |
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForSupport</a>              | AWS Service: support (Service-Linked Role)              | -              |
| <input type="checkbox"/>            | <a href="#">AWSServiceRoleForTrustedAdvisor</a>       | AWS Service: trustedadvisor (Service-Linked Role)       | -              |
| <input type="checkbox"/>            | <a href="#">DynamodbFullAccessrole</a>                | AWS Service: ec2                                        | 55 days ago    |
| <input checked="" type="checkbox"/> | <a href="#">ec2-shutdown-role</a>                     | AWS Service: lambda                                     | -              |
| <input type="checkbox"/>            | <a href="#">Ec2dynamodbFullAccessrole</a>             | AWS Service: ec2                                        | 55 days ago    |
| <input type="checkbox"/>            | <a href="#">Ec2ElasticBeanstalkRole</a>               | AWS Service: ec2                                        | 58 days ago    |
| <input type="checkbox"/>            | <a href="#">EC2IAMFullAccessRole</a>                  | AWS Service: ec2                                        | 68 days ago    |
| <input type="checkbox"/>            | <a href="#">rds-monitoring-role</a>                   | AWS Service: monitoring.rds                             | 89 days ago    |
| <input type="checkbox"/>            | <a href="#">volume-backup-1-role-w0qly3wi</a>         | AWS Service: lambda                                     | -              |
| <input type="checkbox"/>            | <a href="#">volume-backup-role-w0qly3wi</a>           | AWS Service: lambda                                     | -              |

## step3 - Create Lambda Function

Create a Lambda function

Runtime: Python 3.10

Attach the IAM role created earlier

The screenshot displays the AWS Lambda console interface. At the top, the AWS logo and navigation menu are visible. The main heading reads "AWS Lambda lets you run code without thinking about servers." Below this, a "Get started" box encourages creating a function. The "How it works" section shows a code editor with a JavaScript example. The console view for the "new-ec2-shutdown" function is shown, including a success message, function overview, and code source. The code source tab is active, showing a Python script in a code editor.

**new-ec2-shutdown**

Function overview

Diagram | Template

new-ec2-shutdown

Layers (0)

+ Add trigger

+ Add destination

Description

Last modified 38 seconds ago

Function ARN: arn:aws:lambda:us-east-1:658394313504:function:new-ec2-shutdown

Function URL

Code | Test | Monitor | Configuration | Aliases | Versions

Code source

Open in Visual Studio Code | Upload from

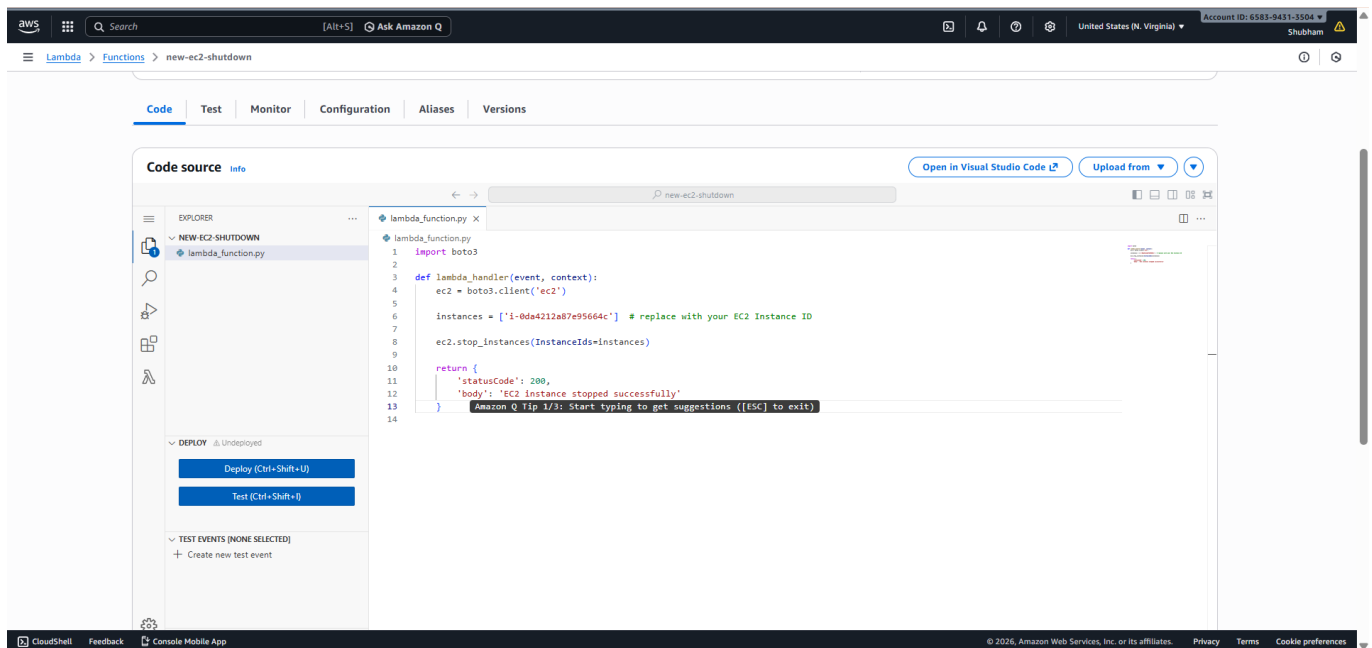
```
1 import json
2
3 def lambda_handler(event, context):
4     # TODO implement
5     return {
6         'statusCode': 200,
```

## step4 - Add Python Code

Add Python code in the Lambda function to:

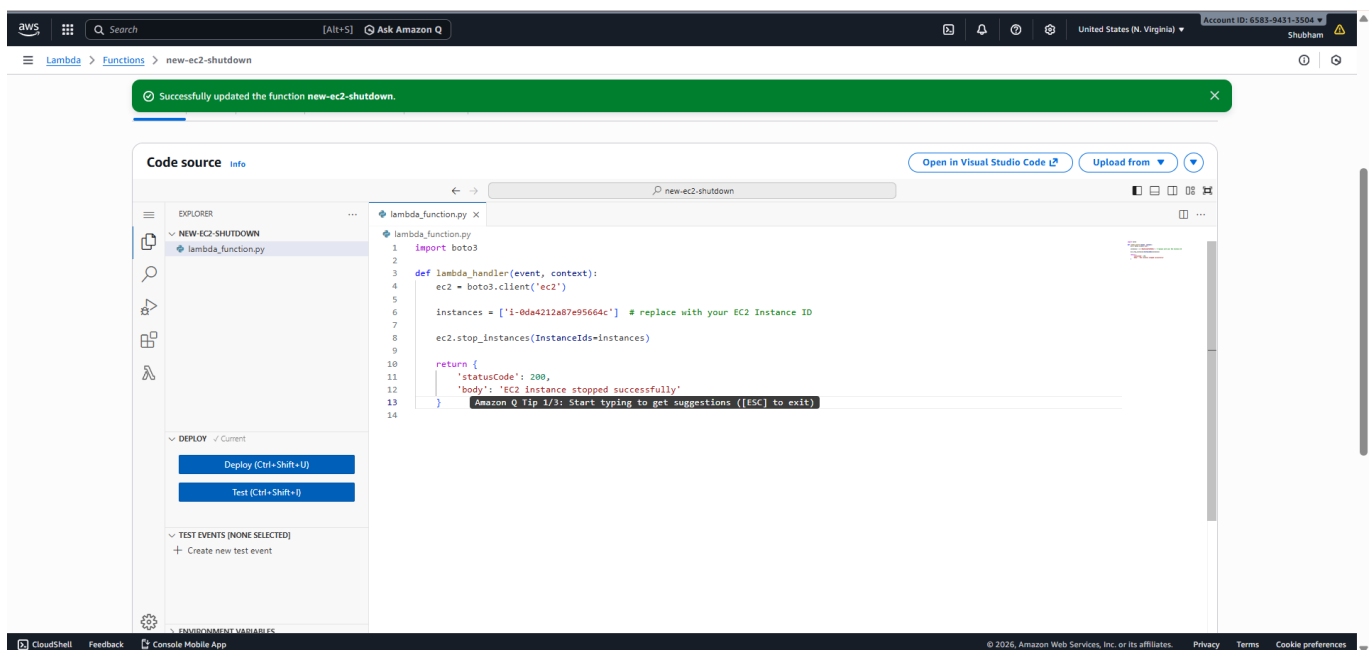
Identify running EC2 instances

Stop those EC2 instances using AWS SDK (boto3)



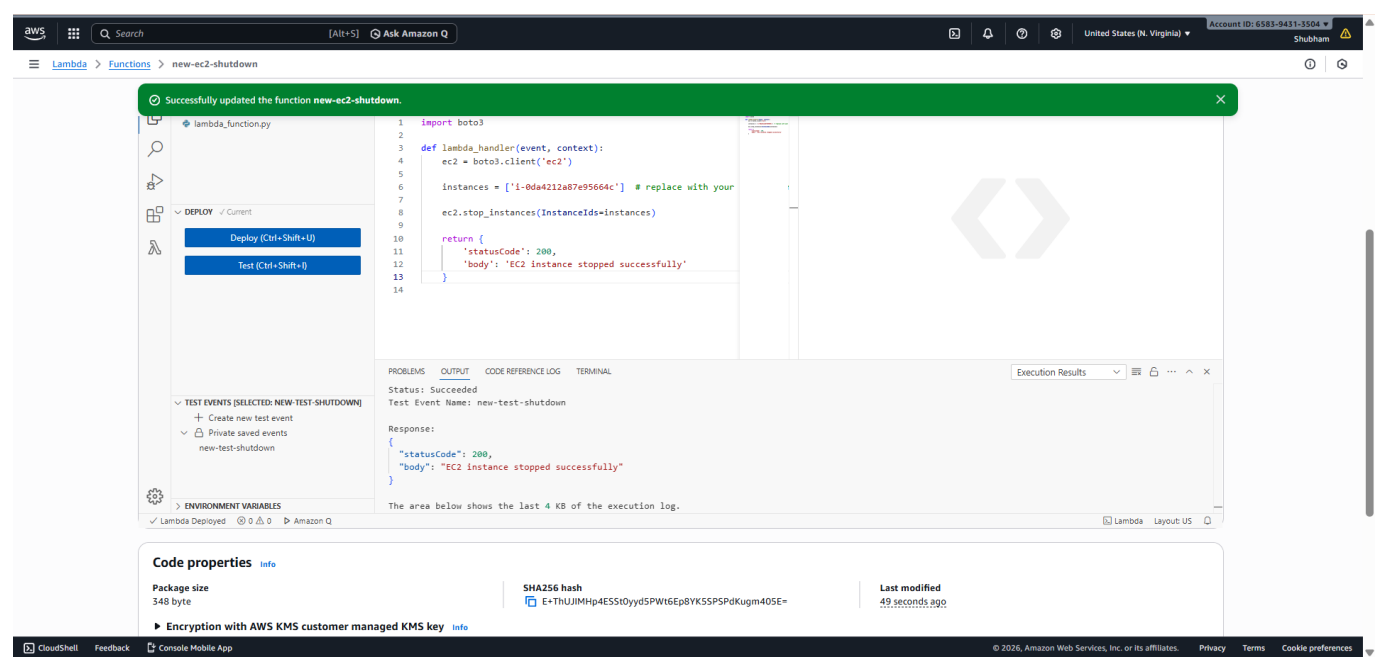
## step5 - Deploy the Function

Deploy the Lambda function after adding the code.



## step6 - Test the Lambda Function

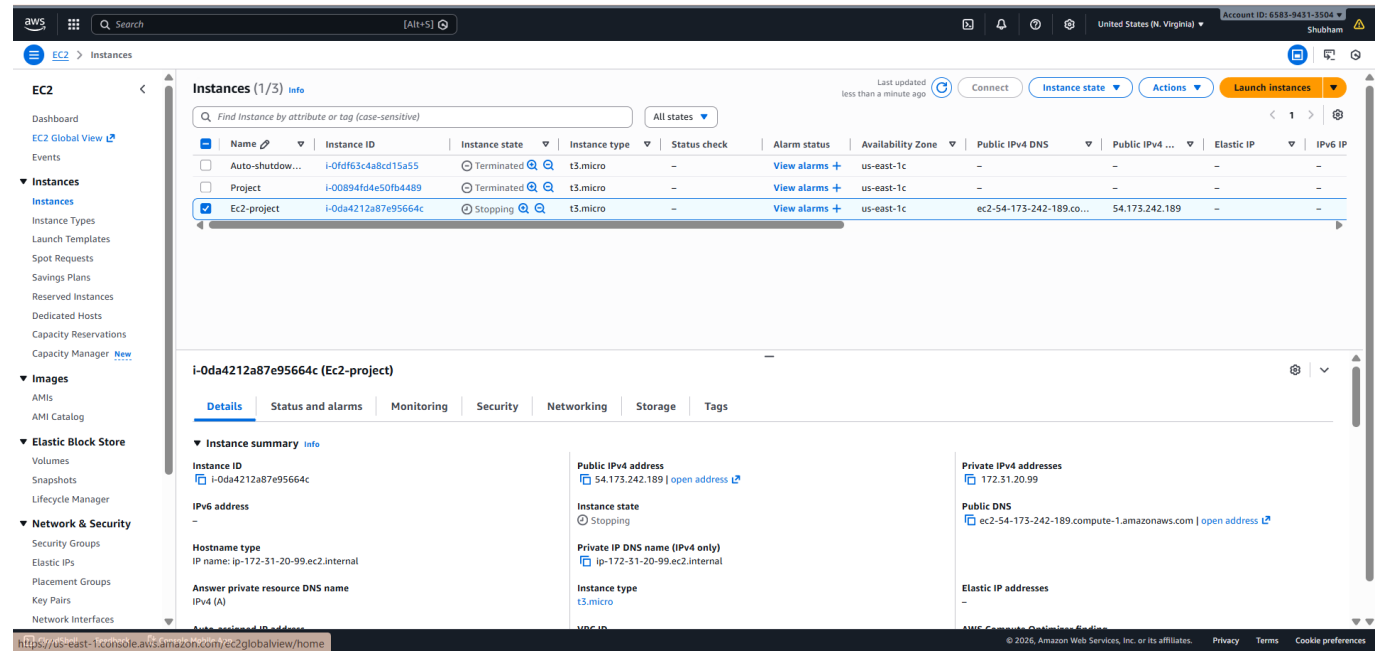
Run a test event from the Lambda console.



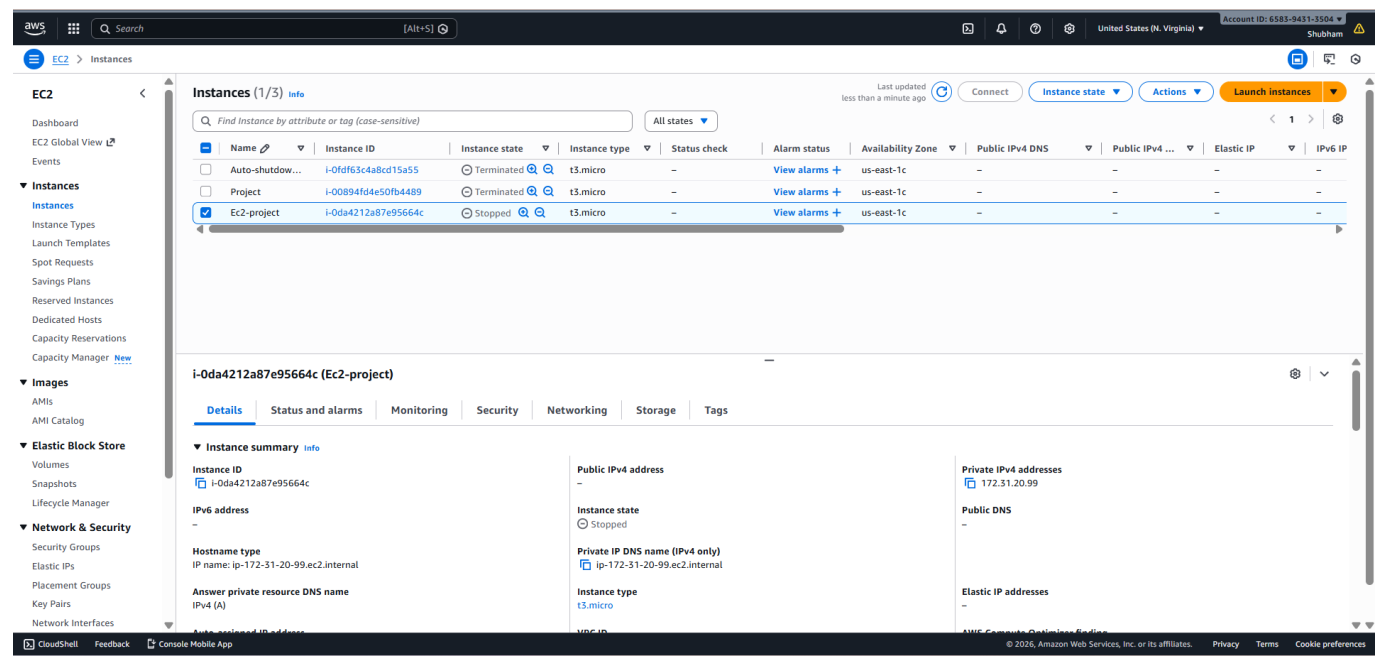
Here, testing of code is succeed.

## step7 - Output

Here, our ec2 instance is stopping.



And finally here ec2 instance is stopped by using aws lambda function.



# Conclusion

This project helped me understand how AWS Lambda can be used to automate EC2 management tasks. By stopping EC2 instances using a Lambda function, I learned how serverless automation can reduce manual effort and help control cloud costs. This project reflects a practical and real-world approach to cloud cost optimization.